Chapter 8

Gold crowns

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CHAPTER CONTENTS

Introduction 103
General considerations 103
Targeted history and examination 103
Sharing of information (for real consent) 104
Indications for full coverage gold crowns 104
Contraindications to full coverage castings 104
Absolute contraindications 104
Steps in tooth preparation for a gold crown 105
General considerations 105
Occlusal design 105
Whether or not to remove previous restorations before preparation for the casting? 106
Where to finish the gold crown preparation with respect to the gingival margin? 106
Burs and instruments 106
Allocation of time between performing basic preparation outline and refining the preparation 107
Has the use of adhesive cements relegated many of the traditional design characteristics to the archive? 107
Tooth preparation for gold crowns 107
Occlusal reduction 107
Axial reduction 110
Preparation/design characteristics specific to a three-quarter gold crown 112
Other castings 114
Making the provisional restoration 114
Impression taking 114
Recording the occlusion 114
Summary 114

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INTRODUCTION

Are gold castings the Cinderella of advanced restorative dentistry? Anecdote would suggest they are excellent restorations; dentists often think that they are the restoration of choice but patients frequently state that the dental aesthetic associated with them is unacceptable. It should be borne in mind that the outcome for gold crowns may appear more favourable because dentists provide such restorations for patients they consider would benefit most; that is, patients are highly selected.

So why should dentists be competent in providing castings, particularly as the provision of densely sintered ceramic restorations are continually being refined? As health care professionals there is an imperative to discuss with the patient the advantages and disadvantages of all restorative options. Indeed, not doing so may result in consent being invalid. A carer's role is to empower the patient such that they can decide how they wish their dental treatment to be advanced.

It is traditional to consider full gold crowns, three-quarter crowns or variations of such, and overlays separately. However, there is commonality between the provisions of such restorations. Such shared characteristics will be stated, but when there are differences, these will be identified. This chapter will exclude gold inlays and onlays which are discussed in Chapter 12.

GENERAL CONSIDERATIONS

Targeted history and examination

These are central to offering the patient what is considered in their best interest. A history should identify the pertinent and ignore the trivial.

Sharing of information (for real consent)

Common to all aspects of treatment, it is important to discuss with the patient their history and examination findings, and to offer different treatment options. The patient should be given time to make an informed choice and contemporaneous clinical notes must be made of this process. If the treatment is of high impact, as is often the case with advanced operative procedures, the decision-making process and the agreed treatment approach should be confirmed in writing.

Indications for full coverage gold crowns

These are listed in order, the first being the most common:

- When a plastic restoration has a history of repeated failure within a short defined time interval (Figure 8.1A). This would include failed adhesive restorations used in the management of tooth wear.
- When difficulty has been experienced in placing a large direct restoration with an adequate contour, contact point and occlusal contacts.
- As a retainer for a fixed prosthesis (see Chapter 19).
- To minimize the *real* risk of tooth fracture, for example after endodontic treatment (Figure 8.1B). Many authorities consider that an overlay/onlay may be the most appropriate restoration for a tooth that has been root-treated.
- To include design characteristics to accommodate a metal-based removable prosthesis (Figure 8.2). Such an indication *per se* may be difficult to justify and the decision to crown the tooth will take other factors into consideration, such as its restorative status. The patient must be made aware that any oral health gain must outweigh the removal of tooth tissue and resources required to provide such restorations.



Figure 8.1A The large restoration in the upper left first molar tooth has undergone repeated fracture. To place a direct restoration in this tooth with adequate contour and contact areas, would be clinically demanding.



Figure 8.1B The lower left first molar tooth has reduced structural integrity and, as a consequence, the disto-lingual cusp has fractured. A full gold crown is indicated in order to preserve the remaining core and coronal tooth tissue.

Contraindications to full coverage castings

Contraindications to providing such restorations include a lifestyle which adversely influences oral health; these are relative and can usually be overcome should the patient so wish. This must be supported by evidence of change such as quitting smoking, modifying the use of erosive drinks, dietary changes to reduce the frequency of sugar consumption or improved home care. Other 'dental' contraindications such as 'active' caries, 'active' periodontal and periradicular disease have been discussed in Chapters 1–3.

A targeted preventative and preparatory phase is at the heart of a treatment plan which includes the provision of successful laboratory fabricated restorations. If this has not been carried out as part of the treatment plan, apart from the dentist not discharging their moral and statutory covenant/contract, a prosecuting barrister may claim, for example: 'My client would not have consented to this crown if they had been informed beforehand of the subsequent necessity for endodontic therapy...or regenerative periodontal procedures etc.'

Partial coverage castings may be the restoration of choice in certain circumstances. For example, if full coverage preparation removes the bulk of the remaining tooth structure.

Absolute contraindications

These are few. Systemic sclerosis could be such an example as the patient may not be able to open their mouth sufficiently to receive such treatment. Profound xerostomia would also be considered by some to be another absolute



Figure 8.2 Full coverage gold crown on lower right second molar. A rest seat, mesial guide plane and mesio-lingual undercut have been created to optimize the success of the removable cobalt chromium prosthesis.

would also be considered by some to be another absolute contraindication but this has to be balanced against the alternative treatment of repeat restorations and loss of function. Another would be if the patient was fearful (dental anxiety). However, supportive therapy may facilitate them receiving such treatment.

Funding and access to health care should never be a contraindication.

STEPS IN TOOTH PREPARATION FOR A GOLD CROWN

General considerations

Occlusal design

- *Conformative or reorganized occlusion*. The dentist can elect to adopt a conformative approach or to reorganize the occlusion. A conformative approach is when the patient is given the same occlusal configuration after the fitting of the restoration as when they attended for the preparation; the occlusion remains unchanged. A reorganized approach is when the relationship between the maxilla and mandible is changed by the gold crown/prosthesis. This approach is usually adopted when most or all of the occluding teeth are to be restored in complex full mouth rehabilitations. In this section, only a conformative approach will be described.
- *Incisal/canine/anterior guidance*. It may be considered more useful to tease this subject apart by considering disclusion of teeth in excursive movements as opposed to occlusion. If the patient has incisal/anterior guidance, the focus will be on establishing a maximum number of bilateral occlusal contacts in maximum intercuspation (ICP). In excursive movements the posterior teeth will disclude and occlusal contacts between posterior teeth will not pose a problem (see Chapter 6). Some authorities suggest that if the patient has incisal/canine/anterior guidance, bodily side shift of the mandible (Bennett movement) is minimized.
- *Group function*. If the patient demonstrates group function, all excursions must be accommodated by the gold crown. Some consider that refinement of the occlusion will be necessary when fitting the casting. This is because Bennett movement (bodily movement of the mandible) is difficult to accurately transfer to an articulator (see below).
- *The occlusal table*. Traditionally, the occlusal table of the restoration is punctuated by fissures and is highly polished. There is no evidence that creation of an intricate fissure pattern has any beneficial effect on function. This is not, however, to denigrate the gnathological approach that had its zenith with full mouth rehabilitations placed on natural teeth (these are beyond the scope of this chapter). The most logical approach is one based around function. A sandblasted

occlusal surface will pick up articulating paper marks more readily than a polished surface and it is often useful to prescribe this to the dental technician. If the patient prefers a polished surface, however, the casting can be polished once the occlusion has been checked and adjusted if necessary.

Whether or not to remove previous restorations before preparation for the casting?

This will depend to a degree as to why the casting is being prescribed and the predicted future integrity of the restoration. Some would assert it is good practice to remove all previous restorations and bases and then replace them with an adhesive core before preparation for a casting. This would avoid embarrassing loss of the restoration during preparation. Others would consider resources could be more productively spent giving oral health messages and a more pragmatic approach would be to carry out the preparation incorporating the existing restoration(s) as the core. If these remain intact and the dentist's intuition is that they can support an indirect restoration, then refrain from placing another core. Advocates of this approach highlight the fact that trauma to the pulp is cumulative and that each time a restoration is replaced more tooth tissue is lost. The final decision as to whether to replace the existing restoration and place a new core will need to be made on an individual basis in consultation with the patient.

Where to finish the gold crown preparation with respect to the gingival margin?

There is an enduring tension between engaging as much tooth structure as possible and encroaching on the gingival domain. In the former case this is performed for reasons of retention and resistance; however, the disadvantage of encroaching on the gingival tissues is that an environment is created in an important area that is unfavourable for the patient to maintain plaque free.

Unequivocally, all restorations should be finished on tooth tissue. The reasons for finishing on tooth tissue are to remove any ledges created by the core and to restrict potential leakage to only that between the casting and tooth.

Burs and instruments

In this subsection, only selected areas will be discussed. In the UK it is conventional to prepare teeth for crowns and bridges using medium-grit diamond burs (Figure 8.3). This is in contrast to other countries where tungsten carbide burs are more commonly used. There is no clinical evidence to show that preparations cut with one or another type of bur result in restorations with a superior outcome.

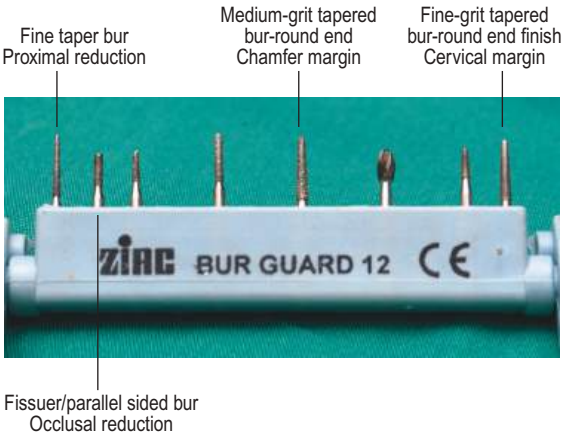


Figure 8.3 Typical crown preparation bur kit. Finer-grit diamond burs (right two) are used to finish the preparation.

Some dentists use fine-grit diamond burs to finish the preparation (Figure 8.3) and argue that the reason for this is to reduce the very small undercuts created by the diamond grit. It has been suggested that these could distort the wax pattern when it is removed from the working die. The routine use of die-spacer on stone dies 'blocks-out' any small undercuts and this should not be an issue (see Chapter 9). In fact the surface roughness of the preparation may improve micromechanical retention when the crown is cemented.

Photographic mirrors are also useful in assessing the axial reduction and taper of the crown preparation. This allows the whole preparation to be examined for possible undercuts without having to change the angle of the mirror or moving the head (Figure 8.4). In addition, a photographic mirror reflects light more effectively.



Figure 8.4 Photographic mirrors are useful in assessing the axial reduction for crown preparations and in assessing the alignment of bridge preparations.

Allocation of time between performing basic preparation outline and refining the preparation

The basic preparation of a tooth can be carried out effectively and safely in a relatively short time. However, a disproportionate amount of time should be spent refining the preparation. It is always important to remember to keep the initial preparation as conservative as possible because more tooth tissue and core can be removed in refinement, but it is not always easy or possible to replace it once it is gone. When carrying out multiple preparations, final refinement may be carried out at a second visit with reference to a stone model. This model is poured from a silicone impression of the preparations after the process of initial preparation has been carried out at the first visit. The most difficult area to prepare and refine is the disto-lingual/palatal angle.

Has the use of adhesive cements relegated many of the traditional design characteristics to the archive?

Some may argue that the slavish following of traditional design characteristics in order to achieve retention and resistance is now not so important. Adhesive cements compensate when traditional characteristics cannot be achieved (Figures 8.5 and 8.6). However, others argue that adhesive cements should only be used when absolutely necessary as the removal of crowns cemented with these can be problematic and time consuming. It is a moot point, but possibly the dentist should not entertain retrievability as these restorations should be there for life. The truth of the matter is that many are not (for a number of reasons) and conventions of crown preparation and cementation should be followed. This also keeps preparations as conservative as possible and in a time when the price of gold is at a high premium the amount of gold used is kept to a minimum; some have referred to gold crowns as gold shell crowns as the crown should consist of a thin even thickness of gold in all aspects.



Figure 8.5 The upper left first molar tooth has been prepared for an adhesively luted three-quarter gold crown. Note the occlusal off-set prepared on the buccal cusps (arrowed) in order to aid location of the restoration during cementation. Apart from this, no occlusal preparation was carried out as 'tooth reduction' had occurred through a combination of a lost restoration and tooth wear.



Figure 8.6 The tooth seen in Figure 8.5 restored with an adhesively luted three-quarter gold casting. Note the satisfactory emergence angle of the restoration achieved by thinning the casting with stones before cementation.

Tooth preparation for gold crowns

Occlusal reduction

Before carrying out occlusal reduction, occlusal contacts when the patient is in intercuspal position (centric stops) and if group function, excursive contacts should be identified using articulating paper. The tooth of the patient illustrated in Figure 8.1B has group function; centric stops and contacts in lateral excursion are identified and differentiated from one another by using different coloured articulating paper.

The occlusal reduction should mirror the anticipated normal morphology of the occlusal table of a tooth. This is best carried out using fissure burs angled according to the cuspal slope (Figure 8.7); this ensures even reduction of the occlusal surface. If the diameter of the bur chosen is consistent with the amount of occlusal reduction needed, the bur can be sunk to its full diameter. If rounded burs are used then there is a potential for insufficient reduction at the centric stops. Not only should sufficient occlusal reduction be carried out to accommodate the gold crown in intercuspal position (~0.5–1 mm) but the dentist should also remove tooth tissue such that there will be no occlusal interferences in all excursive movements including the retruded contact position–intercuspal position (RCP–ICP) slide.



Figure 8.7 To create an occlusal reduction that conforms to the original morphology of the tooth a fissure bur should be used and inclined according to the cuspal slopes. Depth cuts are optional.

When should depth cuts been used?

Some suggest these should always be used as they ensure consistently sufficient and uniform reduction. However, the depth cuts can sometimes be difficult to remove without leaving the preparation with a rippled effect. Figure 8.8 shows a plastic tooth in which depth cuts have been prepared. When removing the islands of tooth tissue or core material between the depth cuts, the fissure bur should be used at a slightly different angle to the initial depth cuts. In this way the bur is prevented from re-entering the depths of the initial cut and forming a rippled occlusal reduction.



Figure 8.8 Depth cuts on the occluso-buccal surface ensure an even and adequate occlusal reduction. The islands of tooth tissue between should be removed with the bur at a different angle (inset).

What is the functional cusp bevel and why should it be prepared?

Once the basic occlusal reduction has been carried out, further reduction over the functional cusp should be undertaken. This is termed the functional cusp bevel and provides sufficient thickness of gold over the functional cusp during mastication and in some patient's parafunction. This gives strength to the restoration where occlusal forces are highest and allows for some degree of wear of the crown with time. So is this necessary if the patient has incisal/canine/anterior guidance and posterior disclusion in lateral excursion? It is controversial as to whether or not those with anterior guidance can exhibit bodily lateral movement of the mandible and the potential for increased wear. The argument is perhaps academic and clinically all that is needed is to ensure sufficient occlusal clearance between the preparation and opposing teeth, both in intercuspal position and excursive movements.

Confirming that sufficient occlusal reduction has been carried out

The most satisfactory way is to look clinically, as this can check out dynamic excursions (Figure 8.9). Another reliable method is to evaluate the thickness of correctly made provisional restorations which have been adjusted to fit the patient's occlusion. This can be achieved by looking for the transmission of light when holding the restoration up at the light or measuring the thickness of the provisional restoration using an Iwanson gauge (see Chapter 9). The other common method is to ask the patient to close against soft wax (Occlusal Indicator Wax, Kerr Hawe) and make excursive movements. This



Figure 8.9 Visual evaluation of occlusal clearance for a full gold crown preparation on the endodontically treated lower right first molar tooth. Sufficient occlusal clearance between the buccal cusp and opposing tooth has been created. Altering the angulation with which the tooth is viewed will allow assessment of clearance lingually.

can be removed from the tooth and examined for excessively thinned areas against light (Figure 8.10). The use of a putty stent taken before tooth preparation is also useful (see Chapter 10). Whatever method is used, if it has been shown that there is inadequate occlusal reduction, the preparation must be refined until sufficient reduction has been achieved.

Should an occlusal offset groove be prepared?

Before this issue is discussed, familiarize yourself with this preparation characteristic (see Figure 8.5). Generally these are not required in full coverage crowns, and where overlays and three-quarter gold crowns are concerned there is no evidence that those restorations incorporating such design characteristics have a superior outcome to those that do not. However, indisputably they permit location and seating of the casting, particularly when using some of the adhesive cements that act as lubricants in their unset state. They also allow increased thickness of gold and strength in otherwise weak areas of the restoration (see later for discussion of three-quarter gold crowns).

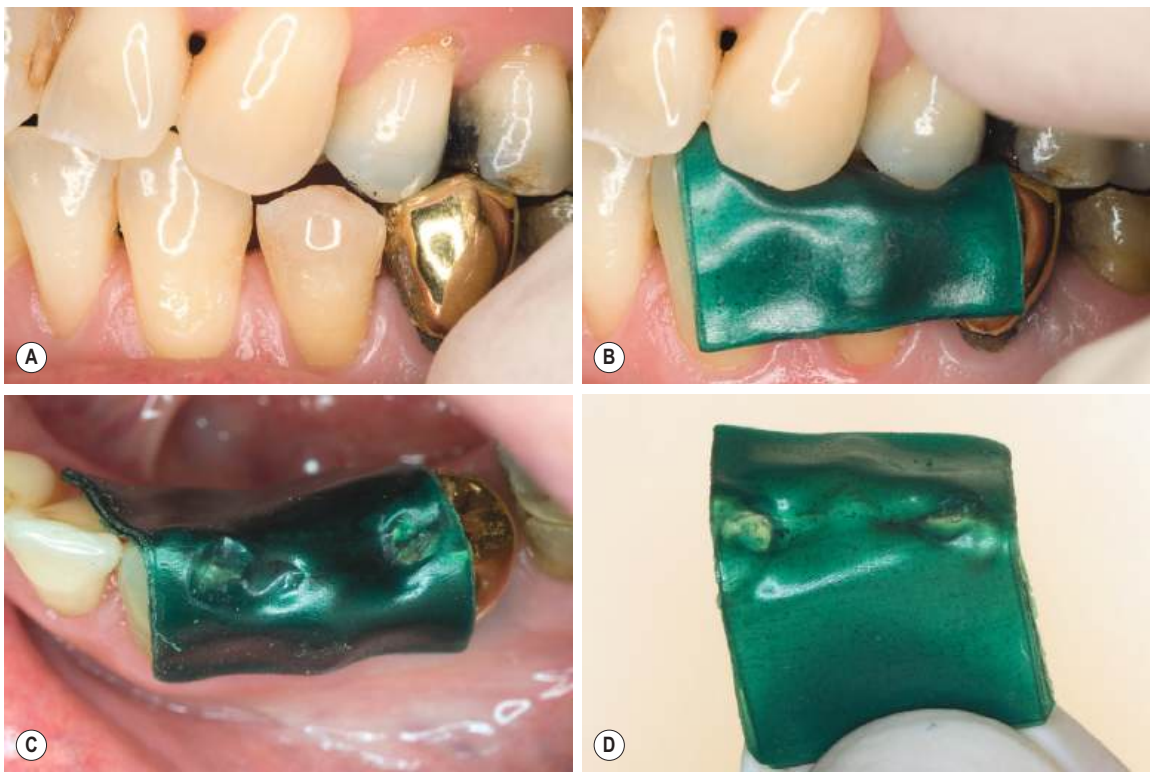


Figure 8.10 Wax used to determine the amount of occlusal clearance for a three-quarter crown preparation on the lower left first premolar tooth (**A**). The wax is 0.5 mm thick and has been doubled over. The patient bites into ICP and excursive movements (**B**). The wax is pierced where there are occlusal contacts on the teeth either side (**C**). The wax should not be thinned where the preparation is; this can be assessed by removing the wax and holding it up to the light (**D**).

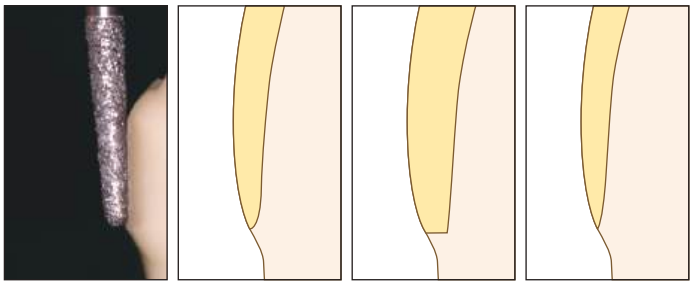


Figure 8.11 The chamfer margin is the most appropriate for gold. It allows for adequate thickness of gold and is easily seen clinically, in the impression and on the stone model in the laboratory. The shoulder margin is too destructive for gold castings and the knife edge margin is difficult to determine. The round ended tapered bur is best for creating a chamfer.



Figure 8.12 Occlusal reduction and buccal and lingual chamfer preparations in the lower left first molar tooth.

Axial reduction

Buccal and lingual preparations (Figures 8.11 and 8.12)

Cervical finishing lines

Traditionally a chamfer is prepared to accommodate the finishing margin and a round-ended tapered bur is ideal for this (Figure 8.11). There is no evidence that restorations with such a design characteristic have a better outcome in the long term. However, it would seem reasonable to use a chamfer as a shoulder would be unnecessarily destructive of remaining tooth structure and a knife-edge finish is not only difficult to identify both clinically and in the laboratory but invariably results in an overbuild of the crown margin and an unfavourable emergence profile. The emergence profile relates to the contour and angle that the tooth (and crown) makes as it emerges from the gingival tissues. An unfavourable emergence profile could lead to gingival trauma if inadequate, and plaque accumulation and potential periodontal problems if too great.

'Parallel belt'

This is in order to create both retention and resistance form. In addition, this characteristic should be of sufficient height to minimize the restoration 'rocking-off' the preparation (resistance).

Retention and resistance form

It is important to consider both the retention and the resistance form of tooth preparation, as it is these that ensure the restoration is not dislodged.

- *Retention form of the preparation*. To understand retention form, two definitions are required: total occlusal convergence and taper. Total occlusal convergence (TOC) is the angle of convergence of two opposing walls (e.g. mesial and distal walls) whilst the taper of a wall is the inclination of one wall in relation to the long axis of the tooth (Figure 8.13). The TOC of the preparation will largely dictate the resistance to dislodgment in an occlusal direction (retention). Surveys of tooth preparations have shown that generally posterior teeth end up with greater TOC than anterior teeth, mandibular teeth have greater TOC than maxillary teeth and bucco-lingual surfaces have a greater TOC than mesio-distal surfaces. These variations may be a direct result of access and vision in various areas of the mouth. The ideal taper has been stated to be in the range of 2–5°, giving a TOC of 4–10°. This is very difficult

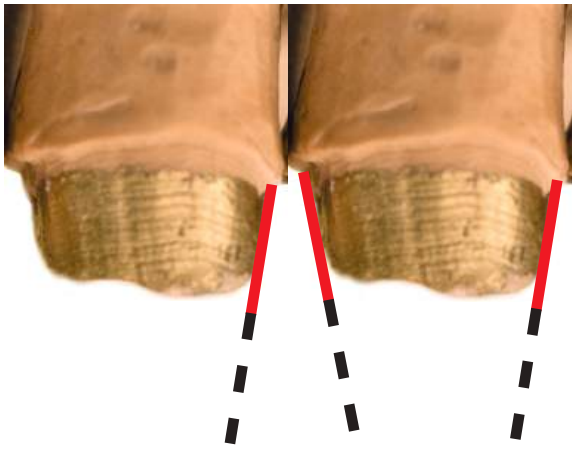


Figure 8.13 Taper, the inclination of one wall to the long axis of the tooth (left) and total occlusal convergence, the angle of convergence of two walls (right).



Figure 8.14 For the axial reduction, visualize the long axis of the tooth (left) and without tilting the bur, move it to cut the buccal (right) and lingual reduction (inset).

to achieve clinically and the more realistic range of 10–20° has been suggested; however, assessment of the TOC has been shown to be greater than this in many instances. Clinically, the TOC is usually assessed by visual means with monocular vision; this often produces preparations with a greater than ideal TOC. At a distance of 30 cm the use of binocular vision can result in areas of the preparation that are undercut not being identified.

- *Resistance form of the preparation*. Resistance to dislodgement in a non-axial direction (resistance) is provided by the TOC, height of preparation (parallel belt) and the ratio of occluso-gingival height to bucco-lingual width. It has been estimated that only 46% of molar preparations have adequate resistance form compared to 96% of incisor preparations. This has been attributed to a general increase in TOC of posterior preparations, the smaller occluso-gingival height of molars and the smaller occluso-gingival to bucco-lingual height ratio.

Where the TOC is inadequate, auxiliary retentive features such as grooves can be prepared. Placement of grooves essentially bisects the TOC, giving greater retention and resistance form. Consider Figure 8.13 for example, placement of a longitudinal groove down the buccal and/or lingual surface would effectively bisect the mesio-distal TOC. If the tooth preparation is more tapered bucco-lingually, the grooves would need to be placed on the mesial and distal surfaces. Much of a crown preparation will involve removal of core material, but if the preparation of grooves would remove sound tooth unnecessarily and weaken the tooth further, consideration should be given to cementing the crown in place with an adhesive luting cement.

To achieve the optimum TOC, a tapered bur should be used and the long axis of the tooth visualized. Keeping the bur in exactly the same plane and resisting the temptation to tilt the bur as the buccal and lingual reduction is carried out will ensure the optimum TOC (Figure 8.14). A useful analogy is as if the handpiece was held by a jig. Ideally the 'parallel belt' or axial preparation should be 3–4 mm in height (Figure 8.15). If its dimension is excessive the restoration would be difficult, if not impossible, to seat.

Parallelism is best gauged by adopting the mantra of: (1) one eye; (2) head still; and (3) the use of a photographic mirror (a hand mirror is easily tilted when viewing single preparations and is always too small to confirm parallelism between preparations when teeth are prepared



Figure 8.15 Full gold crown preparation, lower right first molar tooth. Note the 'ring' formed by the occlusal reduction and chamfer. This would imply a satisfactory 'parallel belt' has been prepared.

for bridges). Parallelism is then confirmed by small movements of the head.

Some, instead of referring to this design characteristic as a 'parallel belt', call this component of the preparation that which accommodates the ferrule. However, in dentistry, a ferrule usually refers to that part of the remaining tooth structure encompassed when providing post crowns (see Chapter 7).

Protecting the soft tissues

The final issue to be highlighted is protection of the soft tissues, particularly a lolling tongue when preparing the lingual cut in the lower posterior sextants. A Svedopter saliva ejector is impregnable whereas disposable flange saliva ejectors (Linguaflex) offer little protection (Figure 8.16). Lingual aspiration with a high volume aspirator tip is the only other way of protecting the tongue when preparing teeth in this sextant. For this you require excellent and reliable four-handed dentistry.

Approximal preparation

The traditional finishing margin for a full gold crown is a chamfer around the complete perimeter of the preparation. This is relatively straightforward to prepare for the buccal and lingual surfaces. However, in order to avoid damage to the adjacent tooth, preparation of an interdental chamfer may be difficult and result in an unacceptable amount of tooth preparation; a knife-edge preparation interdentally may have to be accepted.

As this is the most exacting cut when preparing a tooth for a crown, it will be described in more detail. The most controlled way in order to achieve this is as follows:

- Use direct vision. The patient's head should be moved to the side.
- Cut the knife-edge to the full depth below the contact point, moving the bur from the buccal to the lingual surface in one continuous motion, leaving a thin slither of tooth or core material between the bur and the adjacent teeth (Figure 8.17). This will fall away when the interdental preparation has been completed.
- When 'cutting-out', particularly palatally, the disto-palatal angle is further distally than one anticipates. If prepared more mesially, a destructive groove is made into the core/tooth.
- In order to avoid damage to the adjacent tooth, focus on the adjacent tooth and not the tooth being prepared.



Figure 8.16 The Svedopter (left) and plastic Linguaflex saliva ejector (right and inset) both allow protection of the tongue when preparing lower posterior teeth.

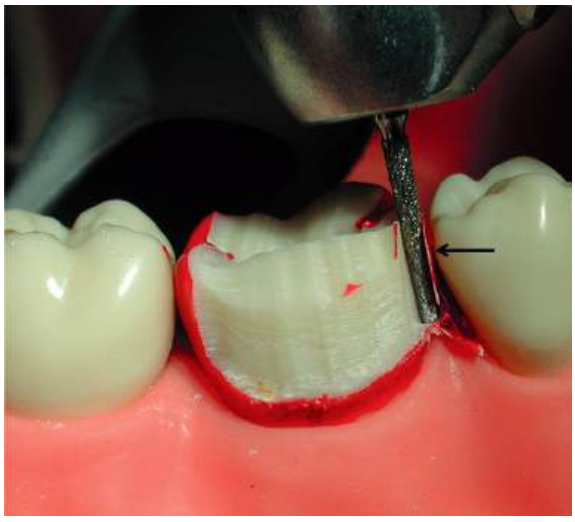


Figure 8.17 The interdental cut is best carried out by a narrow tapered diamond bur leaving a thin slither of tooth between the bur and the adjacent tooth for protection. This will fall away once the cut is complete.

Once the tooth preparation is complete it is important to check that there are no sharp internal line angles and that the margins are smooth and flowing with no steps as the individual aspects to the preparation are merged (Figure 8.18). Sharp internal line angles pose difficulty in the laboratory: it is difficult to pour die stone material into these aspects of the impression and if successful in this, the die stone is liable to wear or chip in these areas, leading to a casting that will not seat fully.

PREPARATION/DESIGN CHARACTERISTICS SPECIFIC TO A THREE-QUARTER GOLD CROWN

As implied earlier in this chapter, it is unclear as to the future role of gold crowns, let alone the three-quarter gold crown. Older members of the profession assert that the



Figure 8.18 Completed full gold crown preparation, lower left first molar tooth. The patient, who is a dentist, did not want the amalgam core replaced with resin composite unless the restoration catastrophically failed during the preparation stage. It is understood the casting that was placed has been satisfactory for more than 10 years.



Figure 8.19 A very old image illustrating a three-quarter gold crown preparation for an upper left first premolar tooth. This preparation was in order to provide a retainer for a conventional cantilever bridge to replace an extracted deciduous upper canine tooth. The restoration met the patient's high dental aesthetic need within the constraints of this traditional treatment approach.

three-quarter gold crown is a superb restoration. However, many patients are unhappy to show any gold, even if it is only a thin line of gold bucco/occlusally. This textbook marks a transition period in indirect restorations with a move toward more aesthetic tooth-coloured restorations. All-ceramic or resin-based restorations are not, however, a panacea solution; in certain circumstances they do not have the physical properties for longevity and tooth preparation for such restorations is often less conservative of tooth tissue. For this reason the three-quarter crown preparation is described here.

It is a myth to suggest that this preparation is the most demanding to prepare – a perfectly executed posterior resin composite is more difficult. Before listing a summary of key design characteristics, the reader is invited to study carefully two illustrations showing this preparation (Figures 8.19 and 8.20).

The design characteristics have evolved from the necessity, in bygone days, to cut grooves in order to provide 'struts' to improve the structural integrity of such restorations where the fourth 'wall' of the crown is missing. The key features are as follows:

- Occlusal and palatal reduction is carried out as for a full coverage crown.
- Both interdental cuts should extend beyond the contact areas in order that the technician can section the resultant model in the laboratory and remove the die to work on.
- The mesial interdental cut should only extend sufficiently to clear the contact area so as not to compromise further the dental aesthetic.
- The distal interdental cut extends buccally in order to afford 'wrap around' to prevent the casting dislodging palatally.



Figure 8.20 Three-quarter gold crown preparation, upper left first molar tooth. A mesial concavity was prepared in the core to accommodate the furcal groove between the mesio-buccal and palatal roots.

- Traditionally, mesial and distal grooves are prepared as far buccally as the interdental cuts will contain.
- Traditionally, an occlusal off-set groove extends between the mesial and distal grooves. This produces an arch of thickened gold buccally, giving the casting strength where the fourth buccal facing to the crown is missing. This preparation characteristic should be adopted judiciously as a compromised cusp could be further weakened.

Figure 8.21 shows the cemented three-quarter crown on the tooth seen in Figure 8.20.



Figure 8.21 Three-quarter gold crown, upper left first molar tooth. Note: (1) satisfactory contact areas; (2) how the palatal margin has been 'thinned' to achieve a satisfactory emergence angle; and (3) polishing of the cervical 2 mm to facilitate optimum home care.

OTHER CASTINGS

Occasionally full coverage castings can have their external surfaces milled and prepared as abutments for a removable or fixed partial prosthesis (Figure 8.22). In the latter situation the bridge can be cemented with temporary cement for ease of removal and maintenance. The problem here is that to allow sufficient thickness of gold to the casting, which is cemented permanently onto the tooth, and then allow space for the overdenture/bridge materials, the tooth preparation is extensive. As a result these are rarely carried out today.

MAKING THE PROVISIONAL RESTORATION

Making a provisional restoration is extremely important. Why and how this is done is described in Chapter 14. In most instances the provisional restoration should be made before the impression is taken. There are two reasons for this:

- The provisional can be used to indirectly assess your tooth preparation. For example, if you are unable to remove the provisional restoration from the preparation or reseat it once fully set, the preparation might be undercut. If the tooth is inadequately prepared in areas, the provisional restoration will be too thin. The tooth preparation can then be modified and a new provisional restoration made if necessary before taking the impression.
- Time management. If you run out of time you have the temporary already made for cementation and the impression can be taken at a subsequent visit.



Figure 8.22 Full coverage telescopic crowns luted permanently, that form abutments for a fixed prosthesis that will be cemented temporarily.

IMPRESSION TAKING

It is pointless to try to take an impression of the preparation if all the margins cannot be identified. Chapter 13 discusses the management of soft tissues to ensure that preparation margins are accessible for taking an impression and it describes the impression materials and techniques used for indirect laboratory-made restorations.

RECORDING THE OCCLUSION

How the occlusion is recorded and whether casts should be mounted on an articulator will depend on how many units of crown or bridges are being placed, the occlusal scheme and the number of remaining teeth. If casts are mounted on an articulator, the type of articulator will be dictated by the complexity of the reconstruction. When there are sufficient index teeth and a single unit is being restored according to a conformative scheme, it is acceptable for the laboratory technician to 'hand-hold' the casts in maximum interdigitation. The reader should refer to Chapter 6 on when and how to articulate casts.

SUMMARY

This chapter has described how teeth are prepared for gold crowns. The subsequent chapter will discuss the materials used in making metal castings, how the crown is made in the laboratory, how the crown is checked for fit in the mouth, how the restoration is cemented and the supportive aftercare.

FURTHER READING

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